MATH 20250
Prof. Lanier

Handout, Day 22

May 8, 2023

Warm-up 1

a) Find two vectors in $\mathbb{R}^3$ that are orthogonal to $\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$

b) Find *all* vectors in $\mathbb{R}^3$ that are orthogonal to $\begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix}$

c) Find two vectors in $\mathbb{R}^3$ that are orthogonal to $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

d)Find *all* vectors in $\mathbb{R}^3$ that are orthogonal to $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

Handout, Day 22

During class 1

The following pair of vectors spans a subspace of $\mathbb{R}^3$. Find an orthonormal basis for this subspace by carrying out the Gram–Schmidt process on this pair of vectors:

$$\begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$$